1.

Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other.

Test Connection Using Ping

Steps:

1. Connect your **laptop and your friend's phone to the same Wi-Fi**.
2. Find the phone's **IP address** from its Wi-Fi/network settings.
3. On your Windows laptop, open **Command Prompt**.
4. Type:

ping 192.168.1.105

(Replace this with your friend's phone's actual IP address.)

5. Press **Enter**.
6. If you see **"Reply from…"**, the devices can communicate.
7. If you see **"Request timed out"**, the devices cannot communicate at that moment.

Result:
"Both devices were connected to the same Wi-Fi. I used the ping command to check communication between my laptop and my friend's phone."

2.

Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.

Microsoft Windows [Version 10.0.26200.9168]

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C:\Users\Sahil>tracert youtube.com

Tracing route to youtube.com [2404:6800:4000:100c::5d]

over a maximum of 30 hops:

1	**2 ms**	**3 ms**	**4 ms**	**2402:3a80:4620:bc7f::c1**
2	**65 ms**	**101 ms**	**101 ms**	**fd00:192:168:174::208**
3	**205 ms**	**103 ms**	**101 ms**	**fd00:192:168:174::208**
4	**91 ms**	**124 ms**	**101 ms**	**fd00:192:168:224::1**
5	**128 ms**	*****	*****	**fd00:192:168:224::2**
6	**97 ms**	**204 ms**	**99 ms**	**2400:5200:c00:64::98**
7				

3.

Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

I disconnected my friend's phone from the Wi-Fi network and then tried to ping it again from my laptop using the command ping 192.168.1.11. The Command Prompt displayed the message *"Request timed out."* This indicates that the laptop could no longer communicate with the phone because it was disconnected from the Wi-Fi network.

4.

Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

Interface: 192.168.235.1 --- 0xb

Internet Address	Physical Address	Type
192.168.235.254	00-50-56-f4-7a-fd	dynamic
192.168.235.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

Interface: 192.168.205.1 --- 0x17

Internet Address	Physical Address	Type
192.168.205.254	00-50-56-f8-08-e9	dynamic
192.168.205.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

Interface: 10.35.218.87 --- 0x18

Internet Address	Physical Address	Type
10.35.218.241	b2-b9-2a-05-ec-13	dynamic
10.35.218.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

5.

Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed.

Simple Network Test Report

Introduction

I connected my laptop and my friend's phone to the same Wi-Fi. Then I checked the network connection using **ping** and **tracert**.

Network Setup

1. I connected my laptop to Wi-Fi.
2. I connected my friend's phone to the same Wi-Fi.
3. I checked the phone's IP address.
4. I used my laptop to test the connection.

Ping Test

I opened **Command Prompt** and entered:

ping [phone IP address]

When the phone was connected, I got **“Reply from…”**. This showed that the two devices could communicate.

Then I disconnected the phone from Wi-Fi and tried again. I got **“Request timed out”** because the phone was disconnected.

Tracert Test

I used this command to check the path to YouTube:

tracert www.youtube.com

The command showed several network **hops** between my laptop and YouTube. The total number of hops was _____.

Results

The ping worked when both devices were connected. After I disconnected the phone, the ping failed and showed **“Request timed out.”** The tracert command showed the different hops to YouTube.

Conclusion

This test helped me understand how a network works. **Ping** checks if a device can be reached, while **tracert** shows the path to a website. I can add screenshots of the command results as proof.